

Problem 2:

Practice Problems on Group Theory

1. Let G be a group of order pq , where p and q are distinct primes. Prove that G is abelian.

Answer: False.

Why: By Sylow theory the Sylow- q subgroup is normal, so G is a semidirect product $P \rtimes Q$.

If $p \nmid (q-1)$ the semidirect product is forced to be direct, but when $p \mid (q-1)$ nontrivial semidirect products exist and can be nonabelian.

Example: $|G| = 6$ gives S_3 , which is nonabelian.

2. Prove that if G is a group of order p^2 , where p is prime, then G is abelian if and only if it has $p+1$ subgroups of order p .

Answer: True.

Why: The only groups of order p^2 are C_{p^2} and $C_p \times C_p$. Thus having $p+1$ subgroups of order p characterizes the abelian $C_p \times C_p$ case.

3. Let G be a finite group and H be a proper subgroup of G . Prove that the union of all conjugates of H cannot be equal to G .

Answer: True.

Why: Let the conjugates be $H_1, \dots, H_k$. Each conjugate intersects another in a proper subset, and counting shows $|\cup_i H_i| \leq k(|H|-1) + 1 \leq [G:H] (|H|-1) + 1 < |G|$. So the union is strictly smaller than G .

4. Let G be a group and N be a normal subgroup of G . If G/N is cyclic, prove that G is abelian.

Answer: False.

Why: Take the dihedral group D_8 (order 8). Its rotation subgroup $N \cong C_4$ is cyclic and normal, and $D_8/N \cong C_2$ is cyclic, yet D_8 is nonabelian. (A correct stronger condition: if N and G/N are cyclic and their orders are coprime, then G is cyclic hence abelian.

5. Prove that in any group G , the set of elements of finite order form a subgroup of G .

Answer: False.

Why: In general nonabelian groups torsion elements need not be closed under multiplication. Example: the infinite dihedral group D_∞ has many reflections of order 2; product of two reflections is a translation of infinite

order. So torsion elements do not form a subgroup in general.

6. Let G be a finite group and p be the smallest prime dividing $|G|$. Prove that any subgroup of index p in G is normal.

Answer: True.

Why: Let H have index p . The action of G on cosets gives $\varphi: G \rightarrow S_p$. By minimality of p the image $\varphi(G)$ must have order either 1 or p ; transitivity forces order p . But a subgroup of S_p of order p fixes a coset, so the kernel of the action equals H . Hence H is normal.

7. Let G be the group and $a, b \in G$. prove that if $a^4 = b^2$ and $ab = ba$, then $(ab)^6 = e$.

Answer: False.

Why: If a and b commute then $(ab)^6 = a^6 b^6$.

From $b^2 = a^4$ we get $b^6 = (b^2)^3 = a^{12}$, so $(ab)^6 = a^{18}$.

There is no reason $a^{18} = e$ in general. Counter-example: in the infinite cyclic group (g) take $a = g$, $b = g^4$. Then $a^4 = g^4 = b^2$, they commute, but $(ab)^6 = g^{18} \neq e$. The claim needs extra hypotheses (e.g. finite orders forcing $a^{18} = e$) to hold.

8. Let G be a group and H be a subgroup of G . Prove that if $[G:H] = n$, then for any $x \in G$, $x^n \in H$.

Answer: False.

Why: The general true statement is $x^n \in H$ for all $x \in G$. Reason: the permutation action

of G on the n cosets gives $\varphi: G \rightarrow S_n$; the order of $\varphi(x)$ divides $n!$ so $x^{n!} \in \ker \varphi = \bigcap_{g \in G} gHg^{-1} \subseteq H$. The exponent n is not sufficient in general.

9. Let G be a finite group and p be a prime number. If G has exactly one subgroup of order p^k for each $k \leq n$, where p^n divides $|G|$. Prove that G has a normal Sylow p -subgroup.

Answer: True.

Why: Let P be the subgroup of order p^n . Any conjugate of P has the same order p^n , hence must equal P by uniqueness. Thus P is normal and is the Sylow p -subgroup.

10. Let G be a finite group and H be a subgroup of G . Prove that if $|G| = p^m$ where p is prime and p does not divide m , and $|H| = p^n$, then H is normal in G .

Answer: True.

Why: A subgroup of order p^n is a Sylow p -subgroup. By Sylow theorems the number n_p of such subgroups divides m and satisfies $n_p \equiv 1 \pmod{p}$. Since $p \nmid m$, the only divisor of m congruent to $1 \pmod{p}$ is 1 so $n_p = 1$. Uniqueness implies normality.